

Summary

Incoming Master of Science in Machine Learning student at the University of Tübingen with a background in empirical machine-learning research and research engineering. My main research interest is deep learning theory, particularly how optimization shapes learning dynamics and the representations neural networks learn. I am moving toward theoretical work while bringing empirical intuition, controlled experimentation, and software engineering experience.

Education

2026– **Master of Science in Machine Learning**, *University of Tübingen*, Tübingen, Germany, *incoming, October 2026*
Research-oriented, four-semester programme. I plan to strengthen my foundations in probability, statistics, optimization, and learning theory.

2025 **Exchange, Computer Science & Engineering**, *Hong Kong University of Science and Technology*, Hong Kong SAR
Courses in machine learning, deep learning in computer vision, large language models, reinforcement learning, and postgraduate data visualisation.

2023–2026 **Bachelor of Science in Engineering, Software Engineering**, *University of Southern Denmark*, Sønderborg, Denmark
Graduated June 2026. GPA: **3.8/4.0 (United States equivalent)**. **Selected as the programme's best bachelor thesis in a graduating cohort of more than 100 students.**

Research Experience

Jan 2026– **External Research Collaborator**, *Hong Kong University of Science and Technology DataVISards*, Hong Kong SAR
Ongoing machine-learning and data-visualisation collaboration established during a 2025 exchange semester.

Jan 2026– **Research Collaborator**, *Applied and Interpretable Machine Learning Research Group*, *University of Southern Denmark*, Sønderborg, Denmark
Initiate and lead research projects with faculty supervision, owning problem formulation, experimental design, implementation, analysis, and manuscript preparation.

2024–2025 **Research Assistant**, *University of Southern Denmark Data & Intelligence Lab*, Sønderborg, Denmark
Built machine-learning pipelines and contributed to study design, literature review, model development, manuscript preparation, grant proposals, and multimodal dataset management.

Selected Publications

2026 **P. M. Konrad**, A.-A. Popa, Y. Sabzehmeidani, L. Zhong, M. Tripathy, A. Constantinescu, E. A. Liehn, S. Ayvaz. *Challenges in Deep Learning-Based Small Organ Segmentation: A Benchmarking Perspective for Medical Research with Limited Datasets*. Accepted, *Biomedical Signal Processing and Control* (Q1, SJR). [arXiv]
Ten-model benchmark with ablations, bootstrap uncertainty, and independent distribution-shift evaluation for small cardiovascular histology datasets.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Self-Reports Do Not Identify Self-Models: An Identifiability Test for Counterfactual Reports*. Oral presentation, *Philosophy of Machine Learning Workshop at the International Conference on Machine Learning*. [OpenReview]
Environment-shift invariance tests across three open models separate report accuracy from mechanism binding.

Selected Publications (continued)

- 2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout*. Poster, *Workshop on Mechanistic Interpretability at the International Conference on Machine Learning*. [OpenReview]
Matched pretrained/instruction-tuned models with probes, activation patching, head ablations, subspace controls, and cross-model replication.
- 2025 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Beyond Major Floods: Deep Learning for Detecting Shallow Water Inundation in Agricultural Areas*. *Knowledge-Based and Intelligent Information & Engineering Systems 2025; Procedia Computer Science 270*, 301–310. [Published version]

Teaching

- 2026 **Teaching Assistant, Artificial Intelligence**, *University of Southern Denmark*
Led exercise sessions, supported students with Artificial Intelligence and Machine Learning concepts, helped design hands-on lab assignments, and developed interactive supplementary course material.

Invited Talks

- Feb 2026 **Agentic Engineering in Practice**, House of Software Sønderborg inaugural event. Invited to introduce agent skills, the Model Context Protocol, and plugin-based development to an audience of more than 100 software engineers, engineering leaders, and founders.

Academic Service

- 2025–2026 **Student Member**, Educational Committee, Software Engineering Programme, University of Southern Denmark. Represented students in curriculum and quality-assurance discussions concerning course content, assessment, and the study environment.

Selected Recognition

- 2026 **Best bachelor thesis**, Software Engineering programme, University of Southern Denmark.
- 2026 **Best Startup Award finalist**, DreamBear, University of Southern Denmark Startup Night; one of five startups selected to pitch.
- 2026 **William Demant Foundation support** for graduate study.
- 2025 **Top 10**, Danish National Championship in Artificial Intelligence; individual competitor, organised by the Danish Society for Artificial Intelligence.
- 2024 **Top 10**, Danish National Championship in Artificial Intelligence; team competitor, organised by the Danish Society for Artificial Intelligence.

Military Service

- 2017–2021 **Staff Duty Soldier**, *Bundeswehr, German Armed Forces*, Glücksburg, Germany
Led a small HR team administering more than 600 soldiers. Received a **Formal Recognition for Exemplary Service** and a **Performance Bonus for Outstanding Achievement** in 2021.

Additional Experience

- 2026– **Founder**, *SaturoLabs*, Denmark
Built and shipped DreamBear, an iOS app creating personalised narrated bedtime stories that frame neurodivergent traits as strengths.
- 2024 **CTO & Co-Founder**, *Tutora ApS*, Sønderborg, Denmark
Led end-to-end development of the company's web application.
- 2022–2024 **Co-CEO & Co-Founder**, *Yeager GmbH*, Remote
Built an Artificial-Intelligence-assisted powerlifting training application.

Technical Skills

- Research Experimental design, controlled interventions, activation patching, representation probing, ablations, bootstrap uncertainty, model training and evaluation, segmentation, conformal prediction, reproducibility.
- Machine Learning Python, PyTorch, Hugging Face Transformers, TransformerLens, scikit-learn, pandas, NumPy, Optuna, Weights & Biases.
- Engineering Git, Docker, Linux, SQL, Google Cloud Platform, JavaScript.

Languages

- Fluent English, German
- Basic Danish, Thai